

**Time:** 4 minutes**Outcome:** Analyze substitution, quality, and new-goods bias.**Difficulty:**

Intermediate

CPI Bias



THE CORE IDEA

The CPI can overstate changes in the cost of living when a fixed basket misses substitution, quality improvements, or new goods. Substitution bias arises when consumers shift toward relatively cheaper alternatives but the fixed basket does not fully reflect that response. CPI remains useful, while substitution, new-good, and quality challenges can make measured inflation differ from true cost-of-living growth. If goods improve in quality, price increases may overstate the rise in the cost of living. If consumers substitute, their actual cost of living may rise less than the fixed basket suggests.



HOW TO RECOGNIZE IT

- If goods improve in quality, price increases may overstate the rise in the cost of living.
- If consumers substitute, their actual cost of living may rise less than the fixed basket suggests.
- A new product gives consumers a useful option that did not previously exist.
- A product becomes more expensive but also much higher quality.
- A laptop price rises while speed and battery life improve.



WATCH OUT

Do not treat all price increases as inflation even when quality or product variety improves. If goods improve in quality, price increases may overstate the rise in the cost of living.



WORKED EXAMPLE: RECOGNIZING CPI BIAS

A fixed basket contains beef even after beef prices rise sharply. Consumers substitute toward chicken, but the CPI continues pricing the original beef quantity. Because the fixed basket misses that cost-saving response, measured inflation can exceed the increase in the cost of maintaining consumers' actual purchasing pattern.



CHECK YOURSELF

A refrigerator sells for the same price but uses much less electricity and lasts longer. Which CPI issue matters?

**READY?**

Return to the game and master this concept.